

Cross-Lingual Mechanistic Interpretability & Safety Alignment for Low-Resource Languages (Gujarati)

1. Project Overview & Scientific Motivation

Large Language Models (LLMs) are primarily aligned and safety-tuned using English-centric datasets. As prompts are translated into low-resource languages such as Gujarati, these safety mechanisms may weaken, enabling harmful requests to bypass alignment safeguards. This phenomenon is known as the cross-lingual safety vulnerability. This project investigates the internal safety representations of Apertus-8B, a fully open multilingual foundation model developed by the Swiss AI Initiative (ETH Zurich, EPFL, and CSCS). The study aims to identify and analyze multilingual refusal vectors, evaluate whether English safety representations transfer to Gujarati, and quantify the Jailbreak Tax, defined as the reduction in model helpfulness caused by stronger safety alignment.

2. Technical Architecture & Specifications

Model: Apertus-8B • Architecture: 8-billion-parameter dense Transformer optimized for multilingual reasoning. • Activation Function: xIELU (Expanded Integral of the Exponential Linear Unit), designed to improve optimization stability and gradient flow in deep multilingual networks. • Compliance: Developed following Swiss transparency principles and EU AI Act requirements, making it suitable for mechanistic interpretability and open-science research.

3. Eight-Week Execution Timeline

Phase 1 (Weeks 1–2): Dataset Preparation & Baseline Probing

- Construct a benchmark containing 100 prompts (50 harmful, 50 benign).
- Translate English prompts into Gujarati using an automated pipeline followed by 20% manual verification to preserve semantic fidelity.
- Validate tokenization, preprocessing, and PyTorch data-loading pipelines.

Phase 2 (Weeks 3–4): Cloud Infrastructure & Activation Extraction

- Provision an NVIDIA L40S or RTX 4090 cloud instance.
- Load Apertus-8B and extract residual stream activations across all 32 Transformer layers for English and Gujarati prompts.
- Develop custom TransformerLens hooks compatible with xIELU while minimizing GPU memory usage and preventing OOM failures.

Phase 3 (Weeks 5–6): Sparse Autoencoder & Causal Steering

- Train a lightweight Sparse Autoencoder (SAE) on layers exhibiting the highest cross-lingual cosine similarity.
- Perform activation patching experiments to test whether injecting English refusal vectors into Gujarati forward passes causally induces safe refusal behavior.

Phase 4 (Weeks 7–8): Evaluation & Research Paper Preparation

- Measure Gujarati jailbreak success rates, translation bypass performance, and the Jailbreak Tax.
- Prepare the final manuscript in LaTeX, emphasizing reproducibility, mechanistic interpretability, multilingual safety, and alignment with the EU AI Act and Swiss AI Initiative open-science practices.